

Consider following grammar: (4 Marks)

$S \rightarrow ASB \mid \Lambda$

$A \rightarrow aAS \mid a$

$B \rightarrow SbS \mid A \mid bb$

i. Eliminate useless symbols, if any.

ii. Eliminate Λ production

Step 1 – Find Useless Symbols

A symbol is useless if:

- it cannot generate terminal strings, OR
- it is not reachable from start symbol

Check Generating Symbols :

Variable A : in the given grammar,

* $A \rightarrow a$

so, A generates terminal

* $B \rightarrow bb$

so, B also generates terminal

* $S \rightarrow \Lambda$

so, S generates terminal

$\therefore$ variable A, B & S are useful

Check Reachability :

from $S \rightarrow ASB$, we can reach A and B

so, all symbols are reachable.

Therefore, No useless symbols.

Step 2: Eliminate Λ productions

Nullable variable : $S \rightarrow \Lambda$

so, S is nullable.

Production 1: $S \rightarrow ASB$

put $S \rightarrow \Lambda$ in RHS

$\therefore S \rightarrow AB$

$\therefore S \rightarrow ASB \mid AB \mid \Lambda$

Production 2 : $A \rightarrow aAS | a$
 $A \rightarrow aAS | aA | a$

Production 3 : $B \rightarrow SbS | A | bb$
 $B \rightarrow SbS | bS | Sb | b | A | bb$

Final Grammar After Eliminating Λ :

$S \rightarrow ASB | AB | \Lambda$
 $A \rightarrow aAS | aA | a$
 $B \rightarrow SbS | bS | Sb | b | A | bb$